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import random

# Part A – Machine Initialization
machines = ['A', 'B', 'C']

# Number of plays
N_play = 10

# Exploration probability
epsilon = 0.2

# Part B – Initialize N, R and Q
N = {'A': 0, 'B': 0, 'C': 0} # Number of times selected
R = {'A': 0, 'B': 0, 'C': 0} # Total reward
Q = {'A': 0, 'B': 0, 'C': 0} # Estimated Q-value

print("Modified E-Greedy Multi-Armed Bandit")
print("-----")

# Part C to Part E – Play 10 times
for play in range(1, N_play + 1):

    # Generate random number
    r = random.random()

    # Modified E-Greedy Decision
    if r < epsilon:
        # Exploration:
        # Select randomly except the machine with highest Q-value
        max_q = max(Q.values())

        highest_machines = [m for m in machines if Q[m] == max_q]

        # If all Q-values are equal, exclude one highest machine
        highest_machine = random.choice(highest_machines)

        exploration_machines = [
            m for m in machines if m != highest_machine
        ]

        selected_machine = random.choice(exploration_machines)

        decision = "Exploration"

    else:
        # Exploitation:
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# Select machine having highest Q-value
max_q = max(Q.values())

best_machines = [m for m in machines if Q[m] == max_q]

selected_machine = random.choice(best_machines)

decision = "Exploitation"

# Part D – Reward Collection
reward = float(input(
    f"Play {play}: Machine {selected_machine} selected ({decision} "
    f"Enter reward: "
))

# Part E – Update N
N[selected_machine] += 1

# Update total reward
R[selected_machine] += reward

# Incremental Q-value update
Q[selected_machine] = (
    Q[selected_machine]
    + (reward - Q[selected_machine]) / N[selected_machine]
)

print(f"Reward received: {reward}")
print(f"Updated Q-value of {selected_machine}: "
      f"{Q[selected_machine]:.2f}")
print()

# Part F – Display Results
print("\n===== FINAL RESULTS =====")

print("\nMachine\tN\tR\tQ-value")

for machine in machines:
    print(
        f"{machine}\t"
        f"{N[machine]}\t"
        f"{R[machine]:.2f}\t"
        f"{Q[machine]:.2f}"
    )

print("\nAnalysis:")
best_machine = max(Q, key=Q.get)

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best_machine = max(Q, key=Q.get)

print(f"Best machine based on Q-value: Machine {best_machine}")
print(f"Highest estimated reward: {Q[best_machine]:.2f}")
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Modified E-Greedy Multi-Armed Bandit

Play 1: Machine B selected (Exploitation). Enter reward: 1
 Reward received: 1.0
 Updated Q-value of B: 1.00

Play 2: Machine B selected (Exploitation). Enter reward: 2
 Reward received: 2.0
 Updated Q-value of B: 1.50

Play 3: Machine B selected (Exploitation). Enter reward: 2
 Reward received: 2.0
 Updated Q-value of B: 1.67

Play 4: Machine B selected (Exploitation). Enter reward: 3
 Reward received: 3.0
 Updated Q-value of B: 2.00

Play 5: Machine A selected (Exploration). Enter reward: 44
 Reward received: 44.0
 Updated Q-value of A: 44.00

Play 6: Machine A selected (Exploitation). Enter reward: 6
 Reward received: 6.0
 Updated Q-value of A: 25.00

Play 7: Machine A selected (Exploitation). Enter reward: 7
 Reward received: 7.0
 Updated Q-value of A: 19.00

Play 8: Machine A selected (Exploitation). Enter reward: 7
 Reward received: 7.0
 Updated Q-value of A: 16.00

Play 9: Machine A selected (Exploitation). Enter reward: 8
 Reward received: 8.0
 Updated Q-value of A: 14.40

Play 10: Machine A selected (Exploitation). Enter reward: 1
 Reward received: 1.0
 Updated Q-value of A: 12.17

===== FINAL RESULTS =====

Machine	N	R	Q-value
A	6	73.00	12.17
B	4	8.00	2.00

C	0	0.00	0.00
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Analysis:

Best machine based on Q-value: Machine A

Highest estimated reward: 12.17